MATH1025B
Essential Mathematical Methods

Chapter 3: Inequalities

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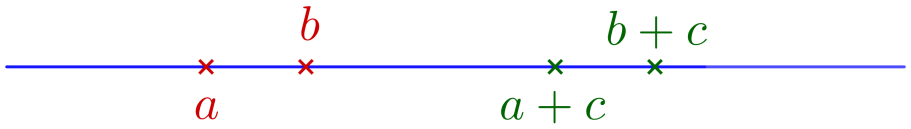
Basic Properties

(i) (Law of Trichotomy) Let $a, b \in \mathbb{R}$. Then exactly one of the following holds:

$$a > b, \qquad a = b, \qquad a < b.$$

(ii) (Transitivity) Let $a, b, c \in \mathbb{R}$. If $a < b$ and $b < c$, then $a < c$.

(iii) Let $a, b, c \in \mathbb{R}$. If $a < b$, then $a + c < b + c$.



(iv) Let $a, b, c \in \mathbb{R}$.

- If $a < b$ and $c > 0$, then $ac < bc$.
- If $a < b$ and $c < 0$, then $ac > bc$.

(v) Let $a, b, r \in \mathbb{R}$. If $0 < a < b$ and $r > 0$, then

- $a^r < b^r$
- $a^{-r} > b^{-r}$ (equivalently, $\dfrac{1}{a^r} > \dfrac{1}{b^r}$).

Example 3.1

Let $x, y > 0$. Prove that $\frac{x}{y} + \frac{y}{x} \geq 2$.

Idea:

$$\frac{x}{y} + \frac{y}{x} \overset{?}{\geq} 2$$
$$\frac{x^2 + y^2}{xy} \overset{?}{\geq} 2$$
$$x^2 + y^2 \overset{?}{\geq} 2xy \quad (\because xy > 0)$$
$$x^2 - 2xy + y^2 \overset{?}{\geq} 0$$
$$(x - y)^2 \overset{?}{\geq} 0 \quad (\text{true!})$$

Proof:

$$(x - y)^2 \geq 0$$
$$x^2 - 2xy + y^2 \geq 0$$
$$x^2 + y^2 \geq 2xy$$
$$\frac{x^2 + y^2}{xy} \geq 2 \quad (\because xy > 0)$$
$$\frac{x}{y} + \frac{y}{x} \geq 2$$

Example 3.2

Let $a, b, c \in \mathbb{R}$. Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ and that the equality holds if and only if $a = b = c$.

Since $(a-b)^2, (b-c)^2, (c-a)^2 \geq 0$, we have

$$\begin{aligned} (a-b)^2 + (b-c)^2 + (c-a)^2 &\geq 0 \\ 2(a^2 + b^2 + c^2 - ab - bc - ca) &\geq 0 \\ a^2 + b^2 + c^2 &\geq ab + bc + ca. \end{aligned}$$

Moreover,

$$\begin{aligned} \text{equality holds} &\iff (a-b)^2 = (b-c)^2 = (c-a)^2 = 0 \\ &\iff a = b = c. \end{aligned}$$

Estimate a fraction $\frac{a}{b}$ where $a, b > 0$

If $p \geq a > 0$ and $b \geq q > 0$, then $aq \leq pq \leq pb$ and so $\frac{a}{b} \leq \frac{p}{q}$.

- *Fixing the numerator, the bigger the denominator, the smaller the fraction.*
- *Fixing the denominator, the bigger the numerator, the bigger the fraction.*

Example 3.3

Suppose that $\frac{x_1}{y_1} < \frac{x_2}{y_2} < \cdots < \frac{x_n}{y_n}$, where $x_i, y_i > 0$ for $i = 1, 2, \ldots, n$, and $n \geq 2$. Show that $\dfrac{x_1}{y_1} < \dfrac{\sum_{i=1}^n x_i}{\sum_{i=1}^n y_i} < \dfrac{x_n}{y_n}$.
Hence, deduce that if $0 < \alpha_1 < \alpha_2 < \cdots < \alpha_n < \frac{\pi}{2}$, then

$$\tan \alpha_1 < \frac{\sum_{i=1}^n \sin \alpha_i}{\sum_{i=1}^n \cos \alpha_i} < \tan \alpha_n.$$

For $i = 1, 2, \ldots, n$, we have $\dfrac{x_1}{y_1} < \dfrac{x_i}{y_i} < \dfrac{x_n}{y_n}$ and thus $\dfrac{x_1}{y_1} \cdot y_i < x_i < \dfrac{x_n}{y_n} \cdot y_i$.
Summing up the inequalities for $i = 1, 2, \ldots, n$, we get

$$\frac{x_1}{y_1} \cdot \sum_{i=1}^n y_i < \sum_{i=1}^n x_i < \frac{x_n}{y_n} \cdot \sum_{i=1}^n y_i,$$

and so

$$\frac{x_1}{y_1} < \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n y_i} < \frac{x_n}{y_n}.$$

Since $0 < \alpha_1 < \alpha_2 < \cdots < \alpha_n < \frac{\pi}{2}$, we have $\tan \alpha_1 < \tan \alpha_2 < \cdots < \tan \alpha_n$.
The result follows by letting $x_i = \sin \alpha_i$ and $y_i = \cos \alpha_i$.

Example 3.3

Show that $P(n) =$ "$\dfrac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} < \dfrac{1}{\sqrt{3n+1}}$" is true for all integers $n \geq 2$.

- When $n = 2$,

$$\begin{aligned} \text{LHS} = \frac{1 \cdot 3}{2 \cdot 4} = \frac{3}{8} &\overset{?}{<} \frac{1}{\sqrt{7}} = \text{RHS} \\ 3\sqrt{7} &\overset{?}{<} 8 \\ 63 = (3\sqrt{7})^2 &\overset{?}{<} 8^2 = 64 \quad \text{(true !)} \end{aligned}$$

Now, we have

$$63 < 64 \implies 3\sqrt{7} = \sqrt{63} < \sqrt{64} = 8 \implies \text{LHS} = \frac{3}{8} < \frac{1}{\sqrt{7}} = \text{RHS}$$

So $P(2)$ is true.

Example 3.4

Show that $P(n) =$ "$\dfrac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} < \dfrac{1}{\sqrt{3n+1}}$" is true for all integers $n \geq 2$.

- Assume $P(k)$ is true where $k \geq 2$.

$$\frac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots 2k(2k+2)} < \frac{1}{\sqrt{3k+1}} \cdot \frac{2k+1}{2k+2} \overset{?}{<} \frac{1}{\sqrt{3k+4}}$$

$$\begin{aligned}
(2k+1)\sqrt{3k+4} &\overset{?}{<} (2k+2)\sqrt{3k+1} \\
(2k+1)^2(3k+4) &\overset{?}{<} (2k+2)^2(3k+1) \\
12k^3 + 28k^2 + 19k + 4 &\overset{?}{<} 12k^3 + 28k + 20k + 4 \\
0 &\overset{?}{<} k
\end{aligned}$$

Indeed $k > 0 \overset{\text{(Exercise!)}}{\Longrightarrow} \cdots \Longrightarrow \dfrac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots 2k(2k+2)} < \dfrac{1}{\sqrt{3k+4}}$.

So $P(k+1)$ is true.

- By mathematical induction, $P(n)$ is true for all integers $n \geq 2$.

Exercise 3.1

(a) Let a and b be two distinct positive real numbers. Show that $a^{n+1} - a^n b > ab^n - b^{n+1}$ for all positive integers n.

(b) By mathematical induction, show that $b^n\left[(n+1)a - bn\right] < a^{n+1}$ for all positive integers n. Hence, prove that $\left(1+\frac{1}{n}\right)^n < \left(1+\frac{1}{n+1}\right)^{n+1}$ for all positive integers n.

Exercise 3.2 (Bernoulli's Inequality)

Prove that, if $x > -1$, then

$$(1+x)^n \geq 1 + nx \quad \text{for all positive integers } n.$$

Solving Inequalities in One Variable

Goal

Finding all possible values of x such that the inequality holds.

Example 3.5

Solve $\dfrac{x+1}{x-1} \leq 2$.

Caution:

$$\frac{x+1}{x-1} \leq 2 \not\Longleftrightarrow x+1 \leq 2(x-1)$$

since $(x-1) < 0$ is possible (when $x < 1$).

Example 3.5

Solve $\dfrac{x+1}{x-1} \leq 2$.

Method 1:

- Case 1: When $x > 1$, $x - 1 > 0$ and

$$\frac{x+1}{x-1} \leq 2 \iff x + 1 \leq 2(x-1) \iff x \geq 3.$$

$\therefore x > 1$ and $x \geq 3$ give $x \geq 3$.

- Case 2: When $x < 1$, $x - 1 < 0$ and

$$\frac{x+1}{x-1} \leq 2 \iff x + 1 \geq 2(x-1) \iff x \leq 3.$$

$\therefore x < 1$ and $x \leq 3$ give $x < 1$.

Answer: $x < 1$ or $x \geq 3$.

Example 3.5

Solve $\dfrac{x+1}{x-1} \leq 2$.

Method 2:

$$\begin{aligned}
\frac{x+1}{x-1} &\leq 2 \\
\frac{x+1}{x-1}(x-1)^2 &\leq 2(x-1)^2 \qquad (\because (x-1)^2 \geq 0) \\
(x+1)(x-1) &\leq 2(x-1)^2 \\
0 &\leq (x-1)(x-3) \\
x \leq 1 \quad &\text{or} \quad x \geq 3.
\end{aligned}$$

However $x \neq 1$.

Answer: $x < 1$ or $x \geq 3$

Example 3.5

Solve $\dfrac{x+1}{x-1} \leq 2$.

Method 3: Note that

$$\frac{x+1}{x-1} \leq 2 \iff 0 \leq 2 - \frac{x+1}{x-1} \iff 0 \leq \frac{x-3}{x-1}.$$

	$x < 1$	$x = 1$	$1 < x < 3$	$x = 3$	$x > 3$
$x-3$	$-$	$-$	$-$	0	$+$
$x-1$	$-$	0	$+$	$+$	$+$
$\dfrac{x-3}{x-1}$	$+$	underfined	$-$	0	$+$

Answer: $x < 1$ or $x \geq 3$

Example 3.6

Solve $x > \dfrac{3}{x} + 2$. (**Answer:** $-1 < x < 0$ or $x > 3$.)

Note that

$$\begin{aligned} x > \frac{3}{x} + 2 &\iff x - \frac{3}{x} - 2 > 0 \\ &\iff \frac{x^2 - 2x - 3}{x} > 0 \\ &\iff \frac{(x-3)(x+1)}{x} > 0. \end{aligned}$$

The solution is $-1 < x < 0$ or $x > 3$.

Exercise 3.3

Solve $\dfrac{(x-2)^3(x-3)}{e^{-x}\sqrt{x-1}} > 0.$ (**Answer:** $1 < x < 2$ or $x > 3$.)

(**Note:** $e^{-x} > 0$ for all $x \in \mathbb{R}$; $\frac{1}{\sqrt{x-1}} > 0$ if $x > 1$, undefined otherwise.)

Absolute Values and Triangle Inequality for Real numbers

Definition 3.1 (Absolute value of a real number)

The **absolute value** of a real number a, denoted by $|a|$, is defined by

$$|a| := \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Geometrically $|a|$ represents the distance between the point 0 and the point a on the real line.

Lemma 3.1

For any $a \in \mathbb{R}$, $|a| = \sqrt{a^2}$. Note: $\sqrt{}$ stands for the non-negative square root.

Proof. Exercise.

Absolute Values and Triangle Inequality for Real numbers

Theorem 3.2

For any $a, b \in \mathbb{R}$, we have

(a) $|-a| = |a|$,

(b) $|ab| = |a||b|$,

(c) $-|a| \leq a \leq |a|$,

(d) *if* $b \geq 0$ *then* $|a| \leq b$ *if and only if* $-b \leq a \leq b$,

(e) *if* $b > 0$ *then* $|a| < b$ *if and only if* $-b < a < b$.

Proof. Exercise.

Theorem 3.3 (Triangle Inequality)

Let $a, b \in \mathbb{R}$. Then $|a+b| \leq |a| + |b|$. Equality holds if and only if $ab \geq 0$.

Proof. Note that

$$|a+b|^2 = (a+b)^2 = a^2 + 2ab + b^2 \leq |a|^2 + 2|a||b| + |b|^2 = (|a| + |b|)^2.$$

Taking square roots yields $|a+b| \leq |a| + |b|$. Moreover, equality holds if and only if $2ab = 2|a||b|$ which is equivalent to $ab \geq 0$. $\square$

Corollary 3.4 (Generalized Triangle Inequality)

Let $n \in \mathbb{Z}^+$. Suppose $a_1, a_2, \ldots, a_n \in \mathbb{R}$. Then

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|.$$

Equality holds if and only if $a_1, a_2, \ldots, a_n$ are all non-negative or all non-positive.

Proof. Straightforward application of mathematical induction.

Corollary 3.5

If $a, b \in \mathbb{R}$, then

(a) $||a| - |b|| \leq |a - b|$,

(b) $|a - b| \leq |a| + |b|$.

Proof.

(a) By Triangle Inequality, $|a| = |(a - b) + b| \leq |a - b| + |b|$.
So $|a| - |b| \leq |a - b|$.
Reversing the roles of a and b gives $|b| - |a| \leq |b - a| = |a - b|$.
Thus, we have $-|a - b| \leq |a| - |b| \leq |a - b|$, and the inequality follows from Theorem 3.2.

(b) $|a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|$. $\square$

Example 3.7

Solve $|x-2|+|2x+1| \geq 4$.

Note that

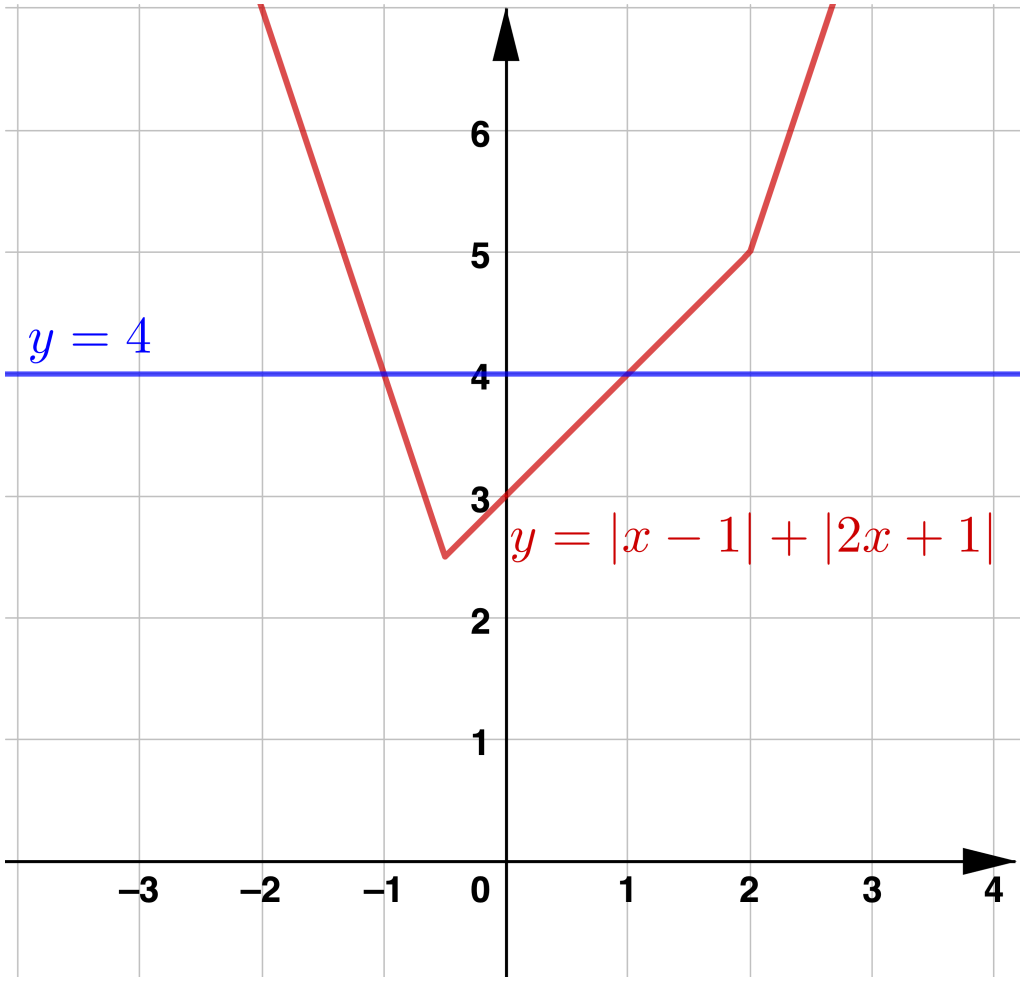
$$|x-2|+|2x+1| = \begin{cases} -(x-2)+[-(2x+1)] = -3x+1 & \text{if } x < -\frac{1}{2} \\ -(x-2)+\quad(2x+1) = x+3 & \text{if } -\frac{1}{2} \leq x < 2 \\ \quad(x-2)+\quad(2x+1) = 3x-1 & \text{if } x \geq 2 \end{cases}.$$

- When $x < -\frac{1}{2}$, $|x-2|+|2x+1| = -3x+1 \geq 4 \iff x \leq -1$; then $x < -\frac{1}{2}$ and $x \leq -1$ gives $x \leq -1$.
- When $-\frac{1}{2} < x \leq 2$, $|x-2|+|2x+1| = x+3 \geq 4 \iff x \leq 1$; then $-\frac{1}{2} < x \leq 2$, and $x \geq 1$ gives $1 \leq x \leq 2$.
- When $x > 2$, $|x-2|+|2x+1| = 3x-1 \geq 4 \iff x \geq \frac{5}{3}$; then $x > 2$ and $x \geq \frac{5}{3}$ gives $x \geq 2$.

Answer: $x \leq -1$ or $x \geq 1$.

Absolute Values and Triangle Inequality for Real numbers

You may plot the graph of $f(x) = |x-2| + |2x+1|$ and solve $f(x) \geq 4$ graphically.



Example 3.8

Solve $|x-2| < x^2$.

Consider two cases.

Case 1: $x < 2$. Then

$$|x-2| < x^2 \iff -(x-2) < x^2 \iff x^2+x-2 > 0 \iff (x+2)(x-1) > 0,$$

which is true when $x < -2$ or $x > 1$. So $x < -2$ or $1 < x < 2$.

Case 2: $x \geq 2$. Then

$$|x-2| < x^2 \iff (x-2) < x^2 \iff x^2 - x + 2 > 0 \iff (x - \frac{1}{2})^2 + \frac{7}{4} > 0,$$

which is always true. So $x \geq 2$.

Combining the two cases, the solution is $x < -2$ or $x > 1$.

Exercise 3.4

Solve $\dfrac{|x+2|}{x+1} < -1$. (**Answer:** $-\frac{3}{2} < x < -1$.)

Example 3.9

Prove that $e^x \geq 1 + x$ (i.e. $e^x - x - 1 \geq 0$) for all $x \in \mathbb{R}$.

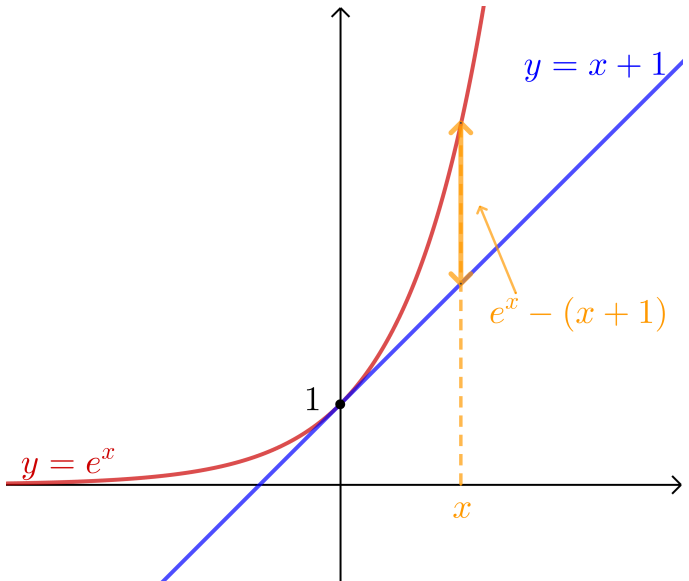
Let $f(x) = e^x - x - 1$. Then $f'(x) = e^x - 1$. So

$$\begin{cases} f'(x) > 0 & \text{if} \quad x > 0 \quad \Longrightarrow \quad f \text{ is strictly increasing on } (0, \infty) \\ f'(x) < 0 & \text{if} \quad x < 0 \quad \Longrightarrow \quad f \text{ is strictly decreasing on } (-\infty, 0). \end{cases}$$

(And f is continuous at $x = 0$.)
Thus f attains its global minimum at $x = 0$, that is,

$$f(x) \geq f(0) = e^0 - 1 = 0 \quad \text{for any } x \in \mathbb{R}.$$

Moreover, the equality holds if and only if $x = 0$.



Exercise 3.5

(a) Show that $f(x) = x^{\frac{1}{3}} - \frac{1}{3}x - \frac{2}{3} \leq 0$ for all $x > 0$.

(b) Using (a), show that $u^{\frac{1}{3}} v^{\frac{2}{3}} \leq \frac{1}{3}u + \frac{2}{3}v$ for all $u, v > 0$.

Exercise 3.6

(a) Suppose $0 < p < 1$.

(i) Let $f(x) = x^p - px + p - 1$ for $x > 0$. Show that $f(x) \leq 0$ for all $x > 0$.

(ii) Let $a, b > 0$. Show that $a^p b^{1-p} \leq pa + (1-p)b$.

(b) Let $\alpha, \beta > 0$. Prove that $a^\alpha b^\beta \leq \left(\dfrac{\alpha a + \beta b}{\alpha + \beta}\right)^{\alpha+\beta}$.

Definition 3.2 (Arithmetic Mean, Geometric Mean)

Let $n \in \mathbb{Z}^+$. Let $a_1, a_2, \ldots, a_n$ be n non-negative real numbers.

(a) The number $\dfrac{a_1 + a_2 + \cdots + a_n}{n}$ is called the **arithmetic mean** of $a_1, a_2, \ldots, a_n$.

(b) The number $\sqrt[n]{a_1 a_2 \cdots a_n}$ is called the **geometric mean** of $a_1, a_2, \ldots, a_n$.

Theorem 3.6 (Arithmetic Mean-Geometric Mean Inequality)

Let $n \in \mathbb{Z}^+$. Let $a_1, a_2, \ldots, a_n$ be n non-negative real numbers. Then

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}.$$

Moreover equality holds if and only if $a_1 = a_2 = \cdots = a_n$.

Lemma 3.7 (Special case of Theorem 3.10: two numbers)

Suppose u, v are non-negative real numbers. Then the inequality $\frac{u+v}{2} \geq \sqrt{uv}$ holds. Moreover equality holds if and only if $u = v$.

Proof. Expanding the inequality $(\sqrt{u} - \sqrt{v})^2 \geq 0$, we have $(\sqrt{u})^2 - 2\sqrt{u}\sqrt{v} + (\sqrt{v})^2 \geq 0$, and hence $\frac{u+v}{2} \geq \sqrt{uv}$. Moreover, equality holds if and only if $\sqrt{u} = \sqrt{v}$, which is equivalent to $u = v$. □

Lemma 3.8 (Special case of Theorem 3.10: 2^n numbers)

Let $n \in \mathbb{Z}^+$. Let $a_1, a_2, \ldots, a_{2^n}$ be 2^n non-negative real numbers. Then the inequality

$$\frac{a_1 + a_2 + \cdots + a_{2^n}}{2^n} \geq \sqrt[2^n]{a_1 a_2 \cdots a_{2^n}}$$

holds. Moreover equality holds if and only if $a_1 = a_2 = \cdots = a_{2^n}$.

Proof of Lemma 3.8.
Denote by $P(n)$ the statement below:
" If $a_1, a_2, \ldots, a_{2^n}$ are 2^n non-negative real numbers, then

$$\frac{a_1 + a_2 + \cdots + a_{2^n}}{2^n} \geq \sqrt[2^n]{a_1 a_2 \cdots a_{2^n}}.$$

Equality holds if and only if $a_1 = a_2 = \cdots = a_{2^n}$."

$P(1)$ is true by Lemma 3.7.

Let $k \in \mathbb{Z}^+$. Suppose $P(k)$ is true. We are going to verify that $P(k+1)$ is true:

Suppose $b_1, \ldots, b_{2^k}, b_{2^k+1}, \ldots, b_{2^{k+1}}$ are 2^{k+1} non-negative real numbers. Since $P(k)$ is true, we have

$$u := \left(\prod_{i=1}^{2^k} b_i\right)^{\frac{1}{2^k}} \leq \frac{1}{2^k}\sum_{i=1}^{2^k} b_i, \qquad v := \left(\prod_{i=2^k+1}^{2^{k+1}} b_i\right)^{\frac{1}{2^k}} \leq \frac{1}{2^k}\sum_{i=2^k+1}^{2^{k+1}} b_i. \quad (*)$$

Moreover $\sqrt{uv} \leq \dfrac{u+v}{2}$ by Lemma 3.7.

Therefore

$$\begin{aligned}
\left(\prod_{i=1}^{2^{k+1}} b_i\right)^{\frac{1}{2^{k+1}}} &= \left[\left(\prod_{i=1}^{2^k} b_i\right)^{\frac{1}{2^k}} \left(\prod_{i=2^k+1}^{2^{k+1}} b_i\right)^{\frac{1}{2^k}}\right]^{\frac{1}{2}} \\
&= \sqrt{uv} \leq \frac{u+v}{2} \qquad (**) \\
&\leq \frac{1}{2}\left[\frac{1}{2^k}\sum_{i=1}^{2^k} b_i + \frac{1}{2^k}\sum_{i=2^k+1}^{2^{k+1}} b_i\right] \\
&= \frac{1}{2^{k+1}}\sum_{i=1}^{2^{k+1}} b_i.
\end{aligned}$$

Furthermore, equality holds if and only if equalities hold in both $(*)$ and $(**)$, which means $u = b_1 = b_2 = \cdots = b_{2^k}$, $v = b_{2^k+1} = b_{2^k+2} = \cdots = b_{2^{k+1}}$ and $u = v$, that is $b_1 = b_2 = \cdots = b_{2^k} = \cdots = b_{2^{k+1}}$.
Hence $P(k+1)$ is true.

By Mathematical Induction, $P(n)$ is true for any $n \in \mathbb{Z}^+$. $\square$

Proof of Theorem 3.10 (AM $\geq$ GM).
Choose $m \in \mathbb{Z}^+ \cup \{0\}$ such that $2^m \leq n < 2^{m+1}$. Set

$$b_1 = a_1, \ \cdots \ , b_n = a_n \quad \text{and} \quad b_{n+1} = b_{n+2} = \cdots = b_{2^{m+1}} = \frac{a_1 + \cdots + a_n}{n}.$$

Then

$$\frac{b_1 + \cdots + b_{2^{m+1}}}{2^{m+1}} = \frac{a_1 + \cdots + a_n + (2^{m+1} - n)\frac{a_1 + \cdots + a_n}{n}}{2^{m+1}} = \frac{a_1 + \cdots + a_n}{n},$$

and

$$b_1 \cdots b_{2^{m+1}} = a_1 \cdots a_n \cdot \left(\frac{a_1 + \cdots + a_n}{n}\right)^{2^{m+1} - n}.$$

By Lemma 3.8, we have

$$\begin{aligned} \frac{a_1 + \cdots + a_n}{n} &= \frac{b_1 + \cdots + b_{2^{m+1}}}{2^{m+1}} \\ &\geq (b_1 \cdots b_{2^{m+1}})^{1/2^{m+1}} = (a_1 \cdots a_n)^{1/2^{m+1}} \cdot \left(\frac{a_1 + \cdots + a_n}{n}\right)^{1 - \frac{n}{2^{m+1}}}, \end{aligned}$$

and thus

$$\frac{a_1 + \cdots + a_n}{n} \geq (a_1 a_2 \cdots a_n)^{1/n}.$$

Finally, equality holds if and only if $b_1 = b_2 = \cdots = b_{2^{m+1}}$, which is equivalent to $a_1 = a_2 = \cdots = a_n$. $\square$

Another proof of AM-GM inequality

Another Proof of Theorem 3.10.
Without loss of generality, we assume that all a_i's are positive.

Let $P(n)$ be the statement
" For any $a_1, a_2, \ldots, a_n > 0$, $\dfrac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}$, and equality holds if and only if $a_1 = a_2 = \cdots = a_n$."

When $n = 1$, LHS $= \frac{a_1}{1} = a_1 =$ RHS. So $P(1)$ is true.

Assume $P(k)$ is true where $k \geq 1$. Let $a_1, a_2, \ldots, a_k, a_{k+1} > 0$.
By the induction assumption, $\dfrac{a_1 + a_2 + \cdots + a_k}{k} \geq \sqrt[n]{a_1 a_2 \cdots a_k}$ and equality holds if and only if $a_1 = a_2 = \cdots = a_k$.
Consider $f(x) := \sum_{i=1}^{k} a_i + x - (k+1)\left[\left(\prod_{i=1}^{k} a_i\right) x\right]^{\frac{1}{k+1}}$ for $x > 0$. Then
$f'(x) = 1 - \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k+1}} x^{-\frac{k}{k+1}}$ and $f'(x) = 0 \iff x = \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}$.

When $0 < x < \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}$, $f'(x) < 0$;

When $x > \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}$, $f'(x) > 0$.

So $f(x) \geq f(\left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}) = \sum_{i=1}^{k} a_i - k\left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}} \geq 0$.

Another proof of AM-GM inequality

Now,

$$\begin{aligned}
\sum_{i=1}^{k+1} a_i - (k+1)\left(\prod_{i=1}^{k+1} a_i\right)^{\frac{1}{k+1}} &= f(a_{k+1}) \geq f\left(\left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}\right) && (1)\\
&= \sum_{i=1}^{k} a_i - k\left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}} \\
&\geq 0 && (2)
\end{aligned}$$

Moreover,

$$\begin{aligned}
\frac{\sum_{i=1}^{k+1} a_i}{k+1} = \left(\prod_{i=1}^{k+1} a_i\right)^{\frac{1}{k+1}} &\iff \text{equalities hold at (1) and (2)} \\
&\iff a_{k+1} = \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k+1}} \text{ and } a_1 = a_2 = \cdots = a_k \\
&\iff a_1 = a_2 = \cdots = a_k = a_{k+1}.
\end{aligned}$$

So $P(k+1)$ is true. By Mathematical Induction, $P(n)$ is true for any $n \in \mathbb{Z}^+$. $\square$

Other proofs of AM-GM inequality

Exercise 3.7

(a) By considering the function $f(x) = x - \ln x - 1$ for $x > 0$, show that $\ln x \leq x - 1$.

(b) Let $a_1, a_2, \ldots, a_n > 0$. Show that $\dfrac{\sum_{i=1}^{n} a_i}{n} \geq \left(\prod_{i=1}^{n} a_i\right)^{\frac{1}{n}}$

Hint: Let $A = \frac{\sum_{i=1}^{n} a_i}{n}$ and put $x = \frac{a_i}{A}$ into the inequality obtained in (a). Sum up the inequalities for $i = 1, 2, \ldots, n$.

Other proofs of AM-GM inequality

Exercise 3.8

(a) Let $x, y > 0$ and let $n \in \mathbb{Z}^+$. Show that $x^{n+1} - (n+1)xy^n + ny^{n+1} \geq 0$.

(b) Let $a_1, a_2, \ldots, a_n > 0$. Let $A_k = \dfrac{1}{k}\sum_{i=1}^{k} a_i$, $G_k = \left(\prod_{i=1}^{k} a_i\right)^{\frac{1}{k}}$ for $k = 1, 2, \ldots, n$. Show that $k(A_k - G_k) \geq (k-1)(A_{k-1} - G_{k-1})$ for $k = 2, 3, \ldots, n$. Hence, show that $A_n \geq G_n$.

Hint: For $n = k-1$, put $x = a_k^{1/k}$ and $y = G_{k-1}^{1/k}$ into the the inequality obtained in (a).

Example 3.10

By AM-GM Inequality, show that

(a) if $a, b, c > 0$, then $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$;

(b) if $n \in \mathbb{Z}^+$, then $(1 + \frac{1}{n+1})^{n+1} > (1 + \frac{1}{n})^n$;

(a) By AM-GM Inequality,

$$\frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{a}}{3} \geq \sqrt[3]{\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}} = 1.$$

(b) Want: $(1 + \frac{1}{n+1}) > \left[(1 + \frac{1}{n})^n\right]^{\frac{1}{n+1}} = [\underbrace{(1 + \frac{1}{n}) \cdots (1 + \frac{1}{n})}_{n \text{ copies}} \cdot 1]^{\frac{1}{n+1}} = \text{GM}.$

Apply AM-GM Inequality to $(1 + \frac{1}{n}), \ldots, (1 + \frac{1}{n}), 1$, we have

$$\text{AM} = \frac{n(1 + \frac{1}{n}) + 1}{n + 1} = 1 + \frac{1}{n + 1} > \text{GM} = \left[(1 + \frac{1}{n})^n\right]^{\frac{1}{n+1}}.$$

Example 3.11

Let $a_1, a_2, \ldots, a_n > 0$. Apply AM $\geq$ GM to $\frac{1}{a_1}, \frac{1}{a_2}, \ldots, \frac{1}{a_n} > 0$, we have

$$\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}} \leq \sqrt[n]{a_1 a_2 \cdots a_n},$$

and equality holds if and only if $a_1 = a_2 = \cdots = a_n$. The number $\dfrac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}$ is called the **harmonic mean** (HM) of a_i's.

Exercise 3.9

Let $a, b, c > 0$. Show that

$$a^3 + b^3 + c^3 \geq a^2b + b^2c + c^2a.$$

Exercise 3.10

Show that $\left(1+\frac{1}{n}\right)^{n+1}$ is strictly decreasing by AM-GM Inequality.

Theorem 3.9 (Cauchy-Schwarz's Inequality)

Let $x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n \in \mathbb{R}$. Then

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right).$$

Moreover the equality holds if and only if there exist $\alpha \in \mathbb{R}$ such that

$$x_i = \alpha y_i \quad \text{for } i = 1, 2, \ldots, n \qquad \text{or} \qquad y_i = \alpha x_i \quad \text{for } i = 1, 2, \ldots, n.$$

Proof.
If $x_1 = \cdots = x_n = 0$, then both sides of the inequality are zero and the inequality is trivially true.
In this case, take $\alpha = 0$, then $x_i = \alpha y_i$ for $i = 1, \ldots, n$.
From now on, assume that $x_1, \ldots, x_n$ are not all zero.

Classical Inequalities: Cauchy-Schwarz's Inequality

Define the function $F : \mathbb{R} \to \mathbb{R}$ by

$$F(t) = \sum_{i=1}^{n} (x_i t - y_i)^2 = \left(\sum_{i=1}^{n} x_i^2 \right) t^2 - 2 \left(\sum_{i=1}^{n} x_i y_i \right) t + \sum_{i=1}^{n} y_i^2 =: At^2 + Bt + C,$$

which is a quadratic function because $A = \sum_{i=1}^{n} x_i^2 > 0$.

Since $F(t) = At^2 + Bt + C \geq 0$ for any $t \in \mathbb{R}$ and $A > 0$, the quadratic equation $F(t) = 0$ has at most one real root.
Therefore, $\Delta = B^2 - 4AC \leq 0$ and we have

$$\left(\sum_{i=1}^{n} x_i y_i \right)^2 = \frac{B^2}{4} \leq AC = \left(\sum_{i=1}^{n} x_i^2 \right) \left(\sum_{i=1}^{n} y_i^2 \right).$$

Furthermore, the equality holds if and only if

$$\begin{aligned} \Delta = 0 \iff & \text{ there exists } \alpha \in \mathbb{R} \text{ such that } 0 = f(\alpha) = \sum_{i=1}^{n} (x_i \alpha - y_i)^2 \\ \iff & \text{ there exists } \alpha \in \mathbb{R} \text{ such that } y = \alpha x_i \text{ for } i = 1, 2, \ldots, n. \quad \square \end{aligned}$$

Another Intepretation of Cauchy-Schwarz's Inequality
Let $\vec{x} = (x_1, x_2, \ldots, x_n)$, $\vec{y} = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$ (with the standard inner product).
Cauchy Schwarz's Inequality says that

$$\left|\sum_{i=1}^{n} x_i y_i\right| \leq \left(\sum_{i=1}^{n} x_i^2\right)^{1/2} \left(\sum_{i=1}^{n} y_i^2\right)^{1/2},$$

that is,

$$|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \, \|\vec{y}\|.$$

Moreover, the equality holds

$$\begin{aligned} &\iff \text{there exists } \alpha \in \mathbb{R} \text{ such that } \vec{x} = \alpha\vec{y} \text{ or } \vec{y} = \alpha\vec{x} \\ &\iff \vec{x} \text{ and } \vec{y} \text{ are parallel.} \end{aligned}$$

Remark: The inequality $-1 \leq \dfrac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \, \|\vec{y}\|} \leq 1$ allows us to define the angle between $\vec{x}$ and $\vec{y}$ to be θ such that $\cos\theta = \dfrac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \, \|\vec{y}\|}$.

Example 3.12

Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$. Apply Cauchy-Schwarz's Inequality to

$$x_i = a_i \quad \text{and} \quad y_i = 1 \quad \text{for } i = 1, 2 \ldots, n,$$

we have

$$\sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}} \geq \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Furthermore, the equality holds if and only if $a_1 = a_2 = \cdots = a_n$.

Remark: The number $\left(\frac{1}{n}\sum_{i=1}^{n} a_i^2\right)^{\frac{1}{2}}$ is called the **root mean square** or **quadratic mean** (QM) of $a_1, a_2, \ldots, a_n > 0$. As a result,

$$\text{QM} \geq \text{AM} \geq \text{GM} \geq \text{HM}.$$

Furthermore, the equalities hold if and only if $a_1 = a_2 = \cdots = a_n$.

Example 3.13

For positive real numbers a, b, c, show that $abc(a+b+c) \leq a^3b + b^3c + c^3a$.

Rewrite the inequality as

$$(a+b+c)^2 \leq \left(a^3b + b^3c + c^3a\right)\left(\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab}\right).$$

Let

$$x_1 = \sqrt{a^3b}, \quad x_2 = \sqrt{b^3c}, \quad x_3 = \sqrt{c^3a},$$

and

$$y_1 = \frac{1}{\sqrt{ab}}, \quad y_2 = \frac{1}{\sqrt{bc}}, \quad y_3 = \frac{1}{\sqrt{ca}}.$$

Then

$$x_1y_1 = a, \quad x_2y_2 = b, \quad x_3y_3 = c.$$

By Cauchy-Schwarz's Inequality, we have

$$\left(\sqrt{a^3b}^2 + \sqrt{b^3c}^2 + \sqrt{c^3a}^2\right)\left(\frac{1}{\sqrt{ab}^2} + \frac{1}{\sqrt{bc}^2} + \frac{1}{\sqrt{ca}^2}\right) \geq (a+b+c)^2,$$

that is

$$\left(a^3b + b^3c + c^3a\right)\left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}\right) \geq (a+b+c)^2.$$

Exercise 3.11

For any $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and $b_1, b_2, \ldots, b_n > 0$, we have

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \cdots + a_n)^2}{b_1 + b_2 + \cdots + b_n}.$$

Exercise 3.12

Given $\theta_1, \theta_2, \ldots, \theta_n \in [0, \frac{\pi}{2})$, prove that

$$\frac{1}{\cos^2\theta_1} + \frac{1}{\cos^2\theta_2} + \cdots + \frac{1}{\cos^2\theta_n} \geq n + \frac{(\sin\theta_1 + \sin\theta_2 + \cdots + \sin\theta_n)^2}{\cos^2\theta_1 + \cos^2\theta_2 + \cdots + \cos^2\theta_n}.$$

Exercise 3.13

Let $x, y, z, w > 0$ and suppose that $xyzw = 16$. Show that

$$\frac{x^2}{x+y} + \frac{y^2}{y+z} + \frac{z^2}{z+w} + \frac{w^2}{w+x} \geq 4$$

with equality holds if and only if $x = y = z = w = 2$.

More about Absolute Value and Inequality

Recall the Triangle Inequality and its Corollary.

Theorem 3.3 (Triangle Inequality)

Let $a, b \in \mathbb{R}$. Then $|a+b| \leq |a| + |b|$. Equality holds if and only if $ab \geq 0$.

Corollary 3.5

If $a, b \in \mathbb{R}$, then

(a) $||a| - |b|| \leq |a - b|$,

(b) $|a - b| \leq |a| + |b|$.

Example 3.14

Show that

$$|2x-1|+|x| \leq 5 \quad \Longrightarrow \quad |x| \leq 2.$$

By Corollary 3.5,

$$|2x-1| \geq |2x|-|1| = 2|x|-1.$$

Hence,

$$\begin{aligned} |2x-1|+|x| \leq 5 \quad &\Longrightarrow \quad 2|x|-1+|x| \leq 5 \\ &\Longrightarrow \quad 3|x| \leq 6 \\ &\Longrightarrow \quad |x| \leq 2. \end{aligned}$$

We do not need to solve the inequality!

Example 3.15

Show that

$$|x-2|<1 \quad \Longrightarrow \quad |x^2-4|<5.$$

Since

$$x^2-4=(x-2)(x+2),$$

we have

$$\begin{aligned}
|x^2-4| &= |x-2|\cdot|x+2| \\
&< |x+2| && (\because |x-2|<1) \\
&= |(x-2)+4| \\
&\leq |x-2|+4 && \text{(by triangle inequality)} \\
&< 1+4=5.
\end{aligned}$$

Example 3.16

Show that

$$|x-2| < 0.005 \quad \Longrightarrow \quad |x^3 - 8| < 0.1.$$

By the triangle inequality,

$$|x^3 - 8| = |x-2| \cdot |x^2 + 2x + 4| \leq |x-2|(|x|^2 + 2|x| + 4).$$

If $|x-2| < 0.005$, then

$$|x| \leq |x-2| + 2 < 0.005 + 2 < 3,$$

and hence

$$\begin{aligned} |x^3 - 8| &\leq |x-2|(|x|^2 + 2|x| + 4) \\ &< |x-2|(3^2 + 2 \cdot 3 + 4) = 19|x-2| \\ &< 19(0.005) < 0.1. \end{aligned}$$

Example 3.17

Let $\delta \in (0, 1]$. Show that

$$|x - 2| < \delta \quad \Longrightarrow \quad |x^3 - 8| < 20\delta.$$

If $|x - 2| < \delta (\leq 1)$, then

$$|x| \leq |x - 2| + 2 < \delta + 2 \leq 3,$$

and hence

$$\begin{aligned} |x^3 - 8| &\leq |x - 2|(|x|^2 + 2|x| + 4) \\ &< |x - 2|(3^2 + 2 \cdot 3 + 4) = 19|x - 2| \\ &< 20\delta. \end{aligned}$$

Example 3.17 shows that if x is close to 2 then x^3 is close to 8, and, more accurately, if x is within a δ-distance from 2 then x^3 is within a 20δ distance from 8. That is, we can control the closeness of x^3 to 8.

Example 3.18

Given a number $\varepsilon > 0$. Find a number $\delta > 0$ such that

$$|x-2| < \delta \quad \Longrightarrow \quad |x^3 - 8| < \varepsilon.$$

If we take $\delta = \varepsilon/20$ or $\delta < \varepsilon/20$, then

$$|x^3 - 8| < 20\delta \quad \Longrightarrow \quad |x^3 - 8| < \varepsilon.$$

By Example 3.17, the implication

$$|x-2| < \delta \quad \Longrightarrow \quad |x^3 - 8| < 20\delta$$

is true under the additional condition that $\delta \leq 1$. To guarantee $\delta \leq 1$, we may set

$$\delta = \text{minimum of } \varepsilon/20 \text{ and } 1.$$

This δ is a solution of the problem. Note that the choice of δ is not unique. (We can always take a smaller δ.)

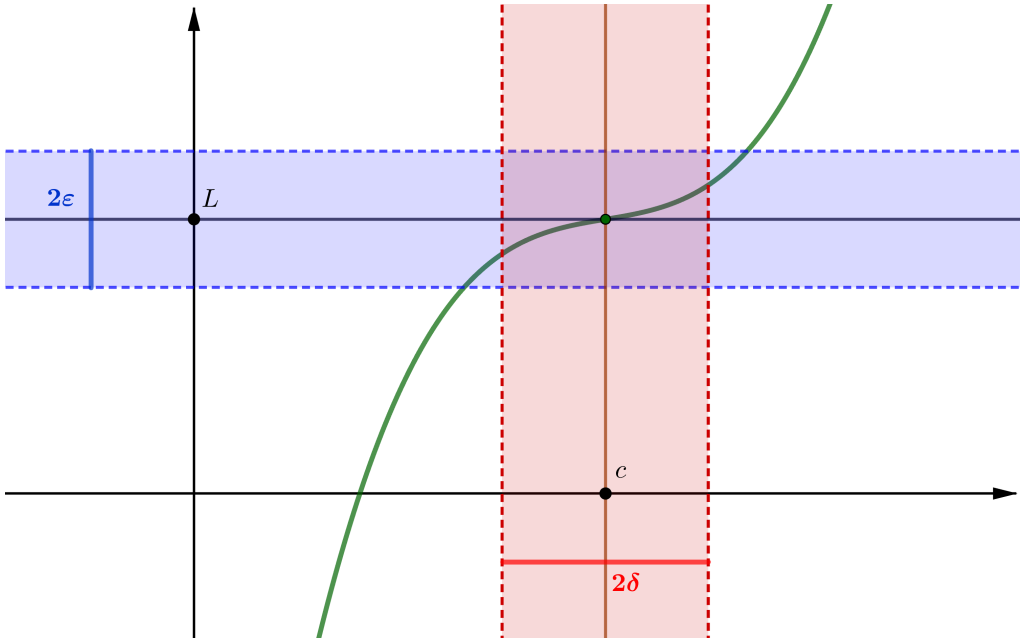
The purpose of Example 3.18 is to motivate the following definition of limit.

Definition 3.3 (Limit of a function at a point)

Let $c \in \mathbb{R}$ and let $f : \mathbb{R}\backslash\{c\} \to \mathbb{R}$ be a function. A real number L is said to a **limit of f at c** (write $\lim_{x \to c} f(x) = L$) if,

given any $\varepsilon > 0$, there exists $\delta > 0$ such that
if $x \neq c$ and $|x - c| < \delta$, then $|f(x) - L| < \varepsilon$.

$$\text{i.e. } (\forall\, \varepsilon > 0)(\exists\, \delta > 0)(\forall\, x \in \mathbb{R}\backslash\{c\})\,[(|x - c| < \delta) \implies (|f(x) - L| < \varepsilon)]$$



GeoGebra Illustration: Epsilon-Delta Definition of a Limit

Exercise 3.14

Let $f(x) = x^2 + 4x$ for $x \in \mathbb{R}$.

(a) Show that if $|x+3| < 1$, then $|f(x)+3| < 3|x+3|$.

(b) Suppose $0 < \varepsilon < 1$. Find one $\delta > 0$ such that

$$|f(x)+3| < \varepsilon \quad \text{whenever } |x+3| < \delta.$$

Example 3.19

Let $p(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$ and $a > 0$. Show that there exists $M > 0$ such that $p(M) > 0$.

Idea:

$$p(M) = \underbrace{aM^3}_{+\text{ve dominant term}} + \underbrace{bM^2 + cM + d}_{\text{possibly } -\text{ve, absorbed by dominant term}}$$

Note

$$\begin{aligned} |bM^2 + cM + d| &\leq |b||M^2| + |c||M| + |d| && \text{(by triangle inequality)} \\ &\leq |b|M^2 + |c|M^2 + |d|M^2 && \text{(if } M \geq 1) \\ &= (|b| + |c| + |d|)M^2. \end{aligned}$$

Then

$$aM^3 + bM^2 + cM + d \geq aM^3 - |bM^2 + cM + d| \geq aM^3 - (|b| + |c| + |d|)M^2 \overset{?}{>} 0,$$

where the last '$>$' is true if $M > \dfrac{|b| + |c| + |d|}{a}$.

Example 3.19

Let $p(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$ and $a > 0$. Show that there exists $M > 0$ such that $p(M) > 0$.

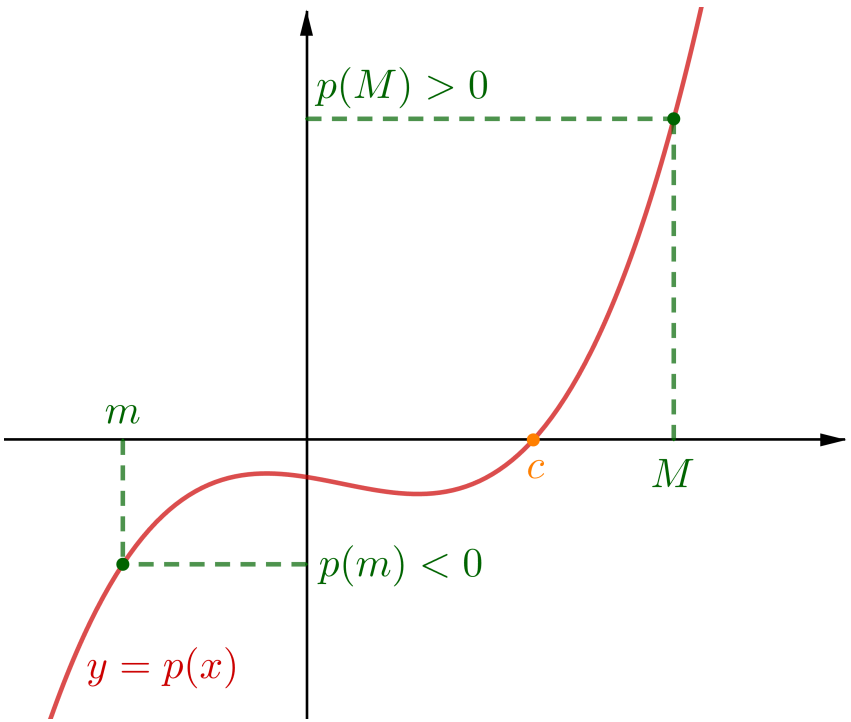
Proof:
Take $M = \dfrac{|b| + |c| + |d|}{a} + 1$.
Then $bM^2 + cM + d \leq (|b| + |c| + |d|)M^2$ since $M \geq 1$.
Hence,

$$\begin{aligned} p(M) &= aM^3 + bM^2 + cM + d \\ &\geq aM^3 - (|b| + |c| + |d|)M^2 \\ &= M^2 \left[aM - (|b| + |c| + |d|)\right] \\ &> 0. \qquad\qquad (\because M > \frac{|b| + |c| + |d|}{a}) \end{aligned}$$

Example 3.19

Let $p(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$ and $a > 0$. Show that there exists $M > 0$ such that $p(M) > 0$.

Remark: Similarly, it can be shown that there exists $m < 0$ such that $p(m) < 0$. By applying the Intermediate Value Theorem to $p(x)$ on $[m, M]$, there exists $c \in (m, M)$ such that $p(c) = 0$. That means that a cubic equation always has a real root.



Exercise 3.15

Under the condition of Example 3.19, show that for any $\alpha > 0$, there exists $M > 0$ such that $p(x) > \alpha$ for all $x > M$. This means that $\lim_{x \to +\infty} p(x) = +\infty$.

Exercise 3.16

Generalize the argument in Example 3.19 to a polynomial of odd degree.

Example 3.20

Let $x_n = \dfrac{n^2 - 3}{n^2 - 5n - 1}$ for $n \in \mathbb{Z}^+$.

(a) Find $N \in \mathbb{Z}^+$ such that $|x_n - 1| \leq \frac{14}{n}$ for all $n \geq N$.

(b) Let $\varepsilon_0 = 2^{-1025}$. Find $N \in \mathbb{Z}^+$ such that $|x_n - 1| \leq \varepsilon_0$ for all $n \geq N$.

(a) For $n \in \mathbb{Z}^+$,

$$|x_n - 1| = \left| \frac{5n - 2}{n^2 - 5n - 1} \right| \leq \frac{5n + 2}{|n^2 - 5n - 1|} \leq \frac{7n}{|n^2 - 5n - 1|}$$

and

$$|n^2 - 5n - 1| \geq n^2 - 5n - 1 \geq n^2 - 6n \geq \frac{1}{2}n^2 \qquad \text{if } n \geq 12.$$

Hence, for all $n \geq 12$,

$$|x_n - 1| \leq \frac{7n}{\frac{1}{2}n^2} = \frac{14}{n}.$$

(b) Take $N = 14 \cdot 2^{1025} \geq 12$. Then for all $n \geq N$, we have

$$|x_n - 1| \leq \frac{14}{n} \leq \frac{14}{N} = 2^{1025} = \varepsilon_0.$$